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Inside the Zone

BY ALEXANDER ZAITCHIK
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CHELYABINSK — It was just after four o'clock in the afternoon on September 29, 1957, when the cooling system failed at the Mayak nuclear complex inside the closed military town of Chelyabinsk-65. Two hundred and fifty cubic meters of volatile liquid uranium waste overheated, then combusted. The fireball shot a kilometer into the sky, where the toxic clouds caught wind and drifted northeast, slicing at roughly 45 degrees between the two nearest cities, Chelyabinsk in the south and Yekaterinburg in the north. It was a less lucky wind for the more than 200 villages and settlements under the fallout's fated path. According to state maps of what is officially known as the "East Ural Radioactive Trace Zone," this path stretched 50 km wide and 300 km long.

The blast in Chelyabinsk-65, since renamed Ozersk, was one of the twentieth-century's best-kept secrets. Its full scope known only to a handful of Soviet officials for more than three decades, it was the first major accident of the atomic age. Until Chernobyl melted down, it was also the largest.

A quarter of a million people were irradiated in the days following the September 1957 explosion. Most of them were soon resettled outside the inner trace zone at the state's expense. But not all of them. Fifteen years after the truth about Mayak spilled out, many of those left behind now believe that while the '57 blast was an accident, their subsequent suffering was part of large-scale human radiation experiment. The question hangs over the East Urals Trace Zone like a mist: Why were some villages evacuated, and others not?

For those left behind, exposure continues through radioactive isotopes in the soil and water with decay rates measured in millennia, and through fresh leakage of radioactive material from Ozersk. For their children and grandchildren, the legacy of 1957 was passed on in the womb.

"This is an intergenerational catastrophe," says Vladimir Chouprov, a nuclear specialist with Greenpeace Russia, which helped organize a demonstration commemorating the 50th anniversary of the Mayak blast last week in Chelyabinsk. "We are seeing the second and third generations living amid radioactive contamination, both accidental and systemic."

Compounding the legacy of radiation in some villages, say local activists and victims, is the inadequate response by Mayak and government officials, neither of whom returned phone calls for this article. After decades of lies and indifference, they say, the lies and indifference continue in new forms.

For the residents of Tatarskaya Korabolka, a dying settlement of 600, the lies started 50 years ago last Saturday.

The townsfolk were in the fields collecting a bumper harvest when they heard it: a solid, dull boom to the west. Ground tremors followed, strong enough to crack windows and rattle plates loose from their shelves. The villagers turned and watched in wonder as a black plume rose high above the cloudless horizon, a dozen kilometers away. "Around the smoke it was the color of sunsets," remembers Gulchara Ismagilova, a witness to the blast who was 11 at the time. Veterans of Stalingrad ordered parents to round up their children and seek shelter. Russia's new enemies, they yelled, have brought war to the southern Urals.

Within hours of the distant blast, villagers handling irradiated hay began to fall sick. Even before police arrived wearing futuristic white suits, locals knew something was terribly, Biblically wrong. But they had no idea what. They would only start to put the pieces together after Chernobyl, three decades later.

The Soviet authorities understood immediately the severity and nature of the disaster. When 300 Korabolka residents out of 5,000 died in the immediate aftermath, the village was slated for complete evacuation by the end of the year. But the planned evacuation never occurred—at least not completely. Instead, a strange thing happened. Its Tatar and Russian halves were handed two separate futures: the ethnic Russian side of the village (population 2,300) was evacuated and razed, while the ethnic Tatar side of the village (population 2,700) was not. There is no trace left of Russian Karabolka, only a forest visible from the nearby road.

With only Tatars remaining, the village was renamed Tatarskaya Korabolka. Today, it is an isolated settlement of traditional one and two-room homes with subsistence backyard plots piled high with firewood. There is a small recreation center for the village's few children, and one small shop. It is a dying community in both senses of the word. The cancer rate is five times the national average. Illnesses grouped under the rubric "chronic radiation sickness" are common. Nearby soil samples still show high levels of Strontium-90. Bitterness runs deep.

"We were left here as a medical experiment," says Gulchara Ismagilova, a lifelong Korabolka resident and the town's leading activist. "What other explanation is there?"

The villagers of Korabolka and some other settlements in the radiation zone weren't just kept where they were in 1957—they were put to work. "We were told without any explanation that the crops had been poisoned and must be burned in earthen pits," remembers Ismagilova.

To destroy contaminated objects, the police turned to the most able bodied and healthy of the community: young adults and children. Directed by local authorities in white protective suits, these workers were known as "liquidators" and were tasked with "mitigation work." They burned crops, buried dead animals, cleaned walls, dismantled buildings and scrubbed their bricks clean. Such work continued for almost a year. According to Chelyabinsk state records, some 1,800 schoolchildren worked as liquidators after the '57 blast.

"My schoolmates and I were sent out to the fields to dig holes and burn the harvest and anything else that was aboveground the day of the explosion," says Ismagilova. "They made us destroy our seed crops for the next year. We had food stocks in our barns, but it was a hungry winter."

The police barred villagers from selling their meat in regional markets, but permitted them to eat it themselves, provided they boiled it for two hours first. Such orders indicate that the authorities themselves did not fully understand radiation.

Those assigned to destroy the irradiated crops were given no protective equipment or any reason to think they needed it. Of Ismagilova's 43 schoolmates and fellow liquidators—all born after WWII—only eight are living. Most suffered shortened lives afflicted with chronic radiation disease. Ten of her fellow liquidators went bald during the work, an affliction that the doctors blamed on "insects." Because doctors could not publicly mention radiation as the cause of the strange post-'57 illnesses, they often blamed insects, the flu, or invented conditions such as "Asthenov Vegetative Syndrome"—sometimes known simply as "The Special Disease." Ismagilova herself spent the summer of 1958 in bed with The Special Disease, wracked by diarrhea and high fever. Today, the fiery 61-year old has maintained a crusading spirit despite suffering an expanding liver, problems with weight loss, and worsening osteoporosis.

"We were ignorant and knew nothing of radioactivity. You couldn't see anything, so there seemed nothing to fear," says Ismagilova, whose advocacy organization is named The Little Liquidators.

As a result of a court battle she helped wage in the 90s, those who can prove they did mitigation work in 1957 and 1958 receive from the state 280 rubles a month (\$11). Often the money fails to arrive or people are dropped from the rolls without reason, and the bureaucratic nightmare of getting the payments flowing falls to Ismagilova, who works as a medical assistant in Ozersk, the closed city of 85,000 that houses the Mayak complex. Much of her free time is spent banging against the closed doors of a corrupt regional administration. Recently a large collection of private donations for the people of Tatarskaya Karobolka was sent to a government office in Chelyabinsk, which was supposed to disburse the money to Ismagilova's organization. But the money disappeared soon after it arrived, never to be seen again.

Ismagilova continues her fight to have the town evacuated and resettled, but admits that is unlikely to happen. The governor of Chelyabinsk, Petr Sumin, personally told her in a 2004 meeting that there was no money for resettlement. And anyway, he added, it wasn't his problem. "This is the USSR's fault, not ours," said the governor.

In lieu of resettlement, Ismagilova fights for whatever help she can get.

"The economy here is dead, the radiation destroyed it," she says. "I know there's no future here, but I promised to help these people, to get them some medical help, remuneration, roads. Something while they are still alive."

* * *

Belief that a cruel medical experiment has been perpetrated against villages downwind and downriver from Mayak is also strong in places unaffected by the '57 explosion. This is especially true of the Tatar village of Muslyumovo, 15 kilometers south of Karobolka, which received its first major dose of Mayak radiation long before the cooling system failure of 1957.

Like other villages along the Techa river, which flows east past Ozersk and the Mayak plant, Muslyumovo's tragedy began in 1948, at the dawn of Russia's nuclear era. It was then that the secret Mayak complex was built to produce plutonium for the country's nascent nuclear arsenal. During the first three years of Mayak's existence, the waste problem was solved in the easiest and cheapest possible way: it was simply pumped directly into the nearby Techa. It was not a decision made in ignorance. Stalin personally forbade Igor Kurchatov, the father of the Soviet bomb, from handling uranium in his research. And Mayak officials had seen Gulag laborers spit blood and die quickly from carrying barrels of radioactive waste with their bare hands. Soviet officials knew the stuff was deadly. It was only when 70 percent of the Techa riverside village of Betlino was diagnosed with cancer in 1951 that the dumping stopped.

But the damage to the river was done. The radioactive waste sunk into the riverbed and bled deep into the sediment along the banks. To this day it remains in the water and the grazing grass of local livestock. The long-term effects can be seen in Muslyumovo, the biggest of a handful of villages (with a total population of about 7,000 between them) that for unknown reasons were not evacuated away from the Techa's most radioactive floodplains and riverbanks.

Throughout the southern Urals, Muslyumovo is a pariah town. The most famous bastard mutant of Mayak, it is shudder-inducing shorthand for all of the murky fears evoked by the specter of that insidious ghost, radiation.

"Few boys want to date or marry a girl from here because they think she'll be barren or give birth to a monster," says a lifelong resident named Mohammed. "If a boy falls in love with a Muslyumovo girl, the parents will step in and put an end to the relationship, even demand a divorce." The fear that Muslyumovo girls cannot produce healthy children is fed by the large collection of hideously deformed miscarriages kept in formaldehyde at the local hospital.

The health problems don't stop at birth. Muslyumovo suffers almost three times the national cancer rate, not counting the hundreds of sick residents regularly monitored by oncologists. Tales of chronic radiation sickness can

be heard from anyone you stop on the street. A 2005 state study found the local drinking water to hold, on average, three times the acceptable level of radiation. Geiger counters placed at different points along the riverbank and nearby fields show 1000-2000 micro Rems per hour—or 50 to 100 times higher than normal. The village gets cleaner the farther you move away from the river, but most villagers live on land registering three times the safe level of Cesium-137, according to a Greenpeace study.

The government does not deny the danger. Starting in the early 50s, it forbade swimming, fishing, and grazing in or around the Techa. But its attempts to enforce these restrictions have always been limp, limited to erecting barbed wire along populated stretches of the river and hiring a single police officer to patrol its banks.

Despite these feeble state attempts to keep people from using the river, up to 15 percent of village children still swim in the Techa during summer, and many of Muslyumovo's poor residents continue to eat its fish. Cows and goats still freely graze on the river's radioactive banks.

In a compromise offer to those agitating for resettlement of the village, the state last year offered residents along the river bank a deal: They could receive one million rubles (\$33,000) or a new house in a settlement on the other side of town, to be christened Novy Muslyumovo. The offer does not vary depending on how many families live in the house, or its size.

Despite the village's 80 percent unemployment rate, the construction company contracted to build the houses is using Central Asian migrant labor. Adding injury to insult, the new settlement is sandwiched between the village graveyard and a brownfield where old pipes are shipped in, burned and scraped, and shipped back out for reuse somewhere else. On windless days when the pipe-cleaning is in progress, a blanket of smoke hovers over what is to be Novy Muslyumovo. Most worrying to locals, the new settlement is not located outside the trace zone. Residents in the new homes will face many of the same dangers as they did before. Their animals will graze on the same grass, their children swim in the same river.

One Muslyumovo resident who accepted the million rubles and left town was Mila Kabirova. Born and raised in Muslyumovo, Kabirova now lives in a one-room flat in Chelyabinsk, where she works in a small shop. In her free time she petitions the government and organizes the locals under the banner of Aigul ("crescent flower" in Tatar), an NGO she founded to fight for evacuation and medical assistance. Kabirova's childhood home was recently reduced to a rubble heap on Karl Marx Street, which runs parallel to the Techa, the morbid and slowly flowing motif that has dominated her life.

Kabira was the youngest of seven children. Her father had the futile job of keeping children (including his own) out of the river. When he died of leukemia in 1962, Kabirova's mother was forced to take a job as a collector of soil, water, and mud samples along the river. Although she didn't know it for years, her employer was actually a Mayak-run laboratory using a Chelyabinsk institute as a front. Mila and her brothers and sisters would often help their mother collect the samples, which they stored under their beds and in the kitchen. Today, all but one of Kabirova's six siblings are dead: two by cancer, three by illnesses associated with chronic radiation sickness. Kabirova herself is frequently ill, has achy bones, and is sterile.

"I am more and more convinced that they kept us here as part of a medical experiment," says Kabirova. "Otherwise why not evacuate us like the others. And why do they still monitor us so closely? Even now, they could have moved the [riverbank families] to a clean area, but they are just moving them down the street, to live with the same water supply and the same dangers. They want people here, that's clear. There's even an economic incentive to stay."

The incentive comes in the form of a 200 rubles a month (\$8) credit that Muslyumovo residents receive for living in a "polluted zone."

Kabirova isn't alone in her conviction the state is using her village for research. According to a 2002 survey conducted by Greenpeace Russia, 62 per cent of residents think they are guinea pigs of the state; three-

quarters of the population say Mayak representatives, including its doctors, are not to be trusted.

There is a 50-year history behind this suspicion. Since 1953, state doctors have been taking fresh local corpses and testing the bones for Strontium-90. In 1959, doctors began conducting annual tests of residents' teeth, especially children. In 1974, yearly spot visits were expanded to include tests of human tissue. Samples were taken to the same place as residents suffering chronic radiation illness: the Center of Radiological Medicine in Chelyabinsk. When established in 1952, the center joined the official Mayak lab as an institution with a keen interest in the maladies of irradiated locals. This interest, say Muslyumovo residents, is noticeably unmatched by an interest in telling people the truth about why they are sick.

"We hate the doctors!" says Robert Timeev, a 57-year-old lifelong Muslyumovo resident who grew up swimming in the river. "They test us, then tell us that so many of us are sick because we smoke too much and are marrying our brothers and sisters." This last insult is especially cutting in Tatar culture, which has strict rules regulating marriage between bloodlines, including distant relations.

"I have chronic radiation disease, my sister just died of cancer, both of my daughters have thyroid problems, and my wife Zoya has chronic asthma," says Timeev, standing in front of his house not far from a particularly hot spot on the Techa. Like most residents of Muslyumovo, Timeev looks much older than his years. He lives on his 400-ruble (\$16) pension and 900-ruble (\$36) health disbursement. (If residents can prove they are sick because of radiation, they qualify for a health benefit of 900 rubles a month. This is one of the faces of the World Bank-backed "monetization" of Russian social services enacted over the last few years.)

Muslyumovo's agrarian economy completely collapsed decades ago. "You can work as a policeman or maybe a teacher here, but that's all," says a lifelong resident named Mohammed. "There's no work. Those who can afford it travel to Chelyabinsk."

As in other villages in the radiation zone, Muslyumovo residents did not begin to connect the dots until Chernobyl. For decades, doctors routinely diagnosed serious illnesses as nothing more than the flu and blamed deformities on incest.

"We didn't think about why we were sick. The doctors were the authority and if they said cancer was something else, or that we were in pain for this reason, we believed them," says Robert Timeev. "We somehow had an idea the Techa had something to do with it. We called it 'river disease'."

Even today, say residents, doctors continue to misdiagnose serious illness or downplay the effect of the river. Often cancers will be diagnosed only when it has advanced. "People around here think they don't want to pay us for very long," says one resident. Once the patient has died, the official cause of death will be put down as "blood circulation disorder."

It's not just the lifelong residents who have fallen victim to the radioactive river. Farida Valeeva moved to Muslyumovo from Chelyabinsk in 1986 after marrying a local man. She fell ill the following year at age 26. For years doctors told her she had the flu. But her flu never went away; it wasted her flesh, made her teeth fall out, and twisted her bones into knots. She developed sclerosis throughout her body. Valeeva too resents the testing of residents as so little help is forthcoming. "They'd come every year and test us but wouldn't tell us the results," says Valeeva, who is 47 but looks twenty years older. One of her sons has chronic nosebleeds, another has been placed in a special class for the mentally slow.

"The doctors still tell us that being afraid and suspicious is bad for our health," says Valeeva, sitting on a thin mattress surrounded by pill bottles in her bedroom. "They say 'radiophobia' is as dangerous as the river. But we didn't know about the river when we got sick. Do the sick birds also have radiophobia?"

It's a similar question to the ones asked by the Chernobyl poet Liubov Sirota in her angry poem, "Radiophobia."

*Is this onlyâ€”a fear of radiation?
Perhapsâ€”the dread of betrayal,
cowardice, stupidity, lawlessness?
the shrinking rivers,
poisoned forests,
children born not to survive . . .
Mighty uncles, what have you dished out
beyond bravado on television?
We will not be resigned
The time has come to sort out
What isâ€”radiophobia?*

Sixteen years after Boris Yeltsin visited Muslyumovo and promised help that never came, residents have sorted it out for themselves.

* * *

Special ed classes like the one Farida Valeeva's son has been placed in are increasingly common in villages along the Techa. Members of the third "Mayak generation" can be seen coming of age at the secondary school in Brodokalmok, 35 kilometers downstream from Muslyomovo. At the time of Mayak's construction in the late 40s, mentally handicapped students were a rare occurrence at the school. Today, 80 out of 360 students are categorized as having mental handicaps. Special classes for students with "slow development" have had to be created to accommodate them, straining the school's meager resources.

Then there are the usual physical ailments. Thyroid disease is common among teachers and students. Many students, says the school principal, also have immune system disorders more usually found in urban schools.

"Before the 1950s, we never had these problems," says Valentina Pashnina, a plump red-haired teacher who runs a small natural history museum on the school's second floor. "The government should have resettled us to a clean place. You can't stop animals from grazing on the river. Even though I teach my students about the dangers of the river, they still swim in it because it looks safe to them. They're just children."

Most of Pashnina's natural history museum is devoted to Brodokalmok's history of radiation. The walls in the small windowless room are lined with pictures of local radiation victims-including local liquidator squads-and trace zone maps. There are also student paintings and dioramas of the Techa that feature bright yellow radiation signs.

Below the museum, on the first floor, the school walls still display faded Soviet-era posters explaining how to prepare for and survive a nuclear attack. The irony does not escape the teachers. To learn about the real radiation the students live with every day, they must attend Pashnina's special class on the subject. She also has a PowerPoint presentation she delivers to the community and her fellow teachers. She and her students were once invited to tour the Mayak plant at Ozersk, an experience that left her marveling at the power of modern public relations. "They were so convincing," she says. "I almost believed them that there was nothing to worry about."

But there is a lot to worry about when it comes to pollution from Mayak, say experts familiar with the current activity of the plant. Radiation from the plant is more than just a dark and little-known chapter of Soviet history effecting a rural slice of the southern Urals. For one thing, nuclear work in Ozersk is today busier than ever.

Beginning in 1976, Mayak began to diversify its work by taking in and reprocessing spent nuclear fuel (SNF) from eastern bloc countries. The long-term contracts were carried over after the dissolution of the USSR, and today Mayak produces 140 tons of waste per year from reprocessing SNF from Bulgaria, Hungary, Ukraine, and others, including Germany. The amount of waste from reprocessing now dwarfs the handful of tons created a year by the plant's rump weapons complex.

Only Russia, France and the UK have reprocessing programs, which puts SNF through a complex process that extracts plutonium. According to Greenpeace Russia, as much as half of the Strontium-90 found along the

Techa is likely the result post-'76 leakage from Mayak's canal system for storing SNF waste.

"The canals are susceptible to overflow during heavy rains, which further contaminates the river," says Vladimir Chouprov, a nuclear expert with Greenpeace. "Because of the regular overflows, the Techa is still taking in fresh radiation that shows for miles."

Many, many miles. Mayak radiation is detectable as far north as the mouth of the Arctic ocean, where the waters of the Techa, then part of the larger Ob' river system, meet the sea. Disturbed by the radiation showing up in its Arctic backyard, the government of Norway in 1997 sponsored a study of Mayak's nuclear activities. As summed up by William Langewiesche in his book *The Atomic Bazaar*, the report concluded that the contamination "is even worse than imagined, that the Mayak facilities have spewed at least twice as much dangerous radiation into the environment as have Chernobyl and all the world's atmospheric bomb tests combined, and that underground lobes of radiation are currently migrating from Mayak's waste-storage reservoirs."

This year marks yet another Mayak anniversary that illustrates the dangers of open air liquid waste storage. In the spring of 1967, the area northeast of Ozersk was blanketed by radiation when heavy winds blew radioactive dust from a drying Lake Karachay, at the time a reservoir for liquid waste. Among the worst effected villages was Tatarskaya Korabolka.

As worrying as the leakage problem is for locals and the government of Norway, the plant presents another, potentially far greater danger, one that could affect more than the southern Urals or the cold water ecosystems at the mouth of the Arctic sea. Mayak facilities currently store 50 tons of weapons-grade-plutonium in light-security ground-level warehouses. Another 38 tons of energy plutonium is stored in other parts of the plant, although this is scheduled for transfer to Zheleznogorsk (formerly Krasnoyarsk-26), another closed city further east, next year.

Some of those familiar with the safeguards at Mayak, including planned upgrades being heavily financed by the U.S. government, worry that the sites are susceptible to sabotage, theft, or accident.

"This is a very dangerous situation," says German Lukashin, a former director of nuclear testing and storage at the Institute of Technical Physics at Snezhinsk, aka "The Russian Los Alamos." Lukashin was forced out of the Institute in 1999 for writing repeated letters to his bosses at the Atomic Energy Agency about the dangers and inadequacies of the storage and waste disposal systems at Mayak. Currently a deputy in the Chelyabinsk municipal council, his biggest worry is the mother of all possible Mayak disasters: a terrorist attack on the above-ground weapons-grade plutonium stocks.

"It's not a safe place, like Yucca Mountain," says Lukashin, referring to the proposed American repository for radioactive waste in Nevada. "You can even find detailed pictures of the [Mayak aboveground storage] buildings on Google Imaging."

The U.S. government has given \$360 million through the Cooperative Threat Reduction Program to safeguard most of the material, which is taken from decommissioned warheads. But Lukashin maintains that even the new safeguards are inadequate.

"People must understand, there is enough material sitting there to contaminate the entire northern hemisphere," he says. "History is very unpredictable. We must get that material underground, and quickly."

If Mayak's troubled past tells us anything, it is that the old whistleblower is probably right.

ABOUT THE AUTHOR

Alexander Zaitchik co-founded Freezerbox in 1998. He has reported from more than a dozen countries for publications such as the *International Herald Tribune*, *Bulletin of Atomic Scientists*, *Wired*, the *San Francisco Chronicle*, *The Believer*, and many others. He lives in New York City.

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